

Tema 10. Completitud de Turing.

10.1. Lenguaje While extendido.

EXWHILE es el conjunto de todos los programas en el lenguaje while extendido, $F(\text{EXWHILE})$ es el conjunto de todas las funciones representadas por EXWHILE.

Puede demostrarse que $\text{WHILE} \subseteq \text{EXWHILE}$ and $F(\text{WHILE}) = F(\text{EXWHILE})$.

Ejemplo 10.1.1. funcion factorial en While extendido.

```
Factorial = (1, s)
s :
  (* counting starts from the previous value *)
  X2 := X1 - 1;
  while X2 ≠ 0 do
    (* macros with prefix or infix notation *)
    X1 := Product(X1, X2);
    X2 := X2 - 1
  od
```

Algunas otras estructuras de control de flujo están permitidas:

```
while C do
  s
od

do f(x) times
  s
od

if C then
  s
fi

if C then
  s1
else
  s2
fi
```

C es definido por un predicado WHILE-numerable, y f(x) es una funcion WHILE-calculable.

10.2. Recursividad => While-calculable.

Teorema 10.2.1. $\text{REC} = F(\text{WHILE})$

Teorema 10.2.1. $\text{REC} \subseteq F(\text{WHILE})$

Teorema 10.2.2. $F(\text{WHILE}) \subseteq \text{REC}$

Proposición 10.2.3. $INI \subseteq F(\text{WHILE})$

Proof.

$\theta \in F(\text{WHILE})$

Given $Q = (0, X1 := 0)$,

$F_Q = \theta \Rightarrow \theta \in F(\text{WHILE})$

$\sigma \in F(\text{WHILE})$

Given $Q = (1, X1 := X1 + 1)$,

$F_Q = \sigma \Rightarrow \sigma \in F(\text{WHILE})$

$\pi \subseteq F(\text{WHILE})$

Given $Q = (k, X1 := Xi)$, $k \geq 1, 1 \leq i \leq k$,

$F_Q = \pi_i^k \Rightarrow \pi_i^k \in F(\text{WHILE}) \Rightarrow \pi \subseteq F(\text{WHILE})$

$INI = \pi \cup \{\theta, \sigma\} \Rightarrow INI \subseteq F(\text{WHILE})$

Proposición 10.2.4.

$f = \bar{g}(h_1, \dots, h_m) \wedge g, h_1, \dots, h_m \in F(\text{WHILE}) \Rightarrow f \in F(\text{WHILE})$

Proof.

Given $f = \bar{g}(h_1, \dots, h_m)$,

$g : \mathbb{N}^m \rightarrow \mathbb{N}, \quad F_G = g,$

$h_i : \mathbb{N}^k \rightarrow \mathbb{N}, \quad F_{H_i} = h_i,$

$m \geq 1, k \geq 0$

and

$C = (k, s)$

$s :$

$X_{k+1} := H_1(X_1, \dots, X_k);$

...

$X_{k+m} := H_m(X_1, \dots, X_k);$

$X_1 := G(X_{k+1}, \dots, X_{k+m})$

then $F_C = f \Rightarrow f \in F(\text{WHILE})$

Proposición 10.2.5.

$$f = \langle g|h \rangle \wedge g, h \in F(\text{WHILE}) \Rightarrow f \in F(\text{WHILE})$$

Proof.

Given $f = \langle g|h \rangle$,

$$g : \mathbb{N}^k \rightarrow \mathbb{N}, \quad F_G = g,$$

$$h : \mathbb{N}^{k+2} \rightarrow \mathbb{N}, \quad F_H = h,$$

$$k \geq 0$$

and

$$R = (k+1, s)$$

$s :$

$$X_{k+2} := G(X_1, \dots, X_k);$$

do X_{k+1} **times**

$$X_{k+2} := H(X_1, \dots, X_k, X_{k+3}, X_{k+2});$$

$$X_{k+3} := X_{k+3} + 1$$

od

$$X_1 := X_{k+2}$$

then $F_R = f \Rightarrow f \in F(\text{WHILE})$

Proposición 10.2.6.

$$f = \mu[g] \wedge g \in F(\text{WHILE}) \Rightarrow f \in F(\text{WHILE})$$

Proof.

Given $f = \mu[g]$,

$$g : \mathbb{N}^{k+1} \rightarrow \mathbb{N}, \quad F_G = g,$$

$$k \geq 0$$

and

$$M = (k, s)$$

$s :$

while $G(X_1, \dots, X_k, X_{k+1}) \neq 0$ **do**

$$X_{k+1} := X_{k+1} + 1$$

od

$$X_1 := X_{k+1}$$

then $F_M = f \Rightarrow f \in F(\text{WHILE})$